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A Novel Logic Styles used for Leakage Power Reduction in MOS Integrated Circuit

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Abstract

Modified constant delay logic style and clocked logic style is defined by comparing their result. In this paper, a design technique has been proposed which reduces the power dissipation. The design and implementation of full adder and ripple carry adder with constant delay logic. The leakage power has become a serious concern in CMOS technologies that has been solved by the MCD and clocked logic style. Constant delay logic style is having two blocks and having a unique characteristic at which we get pre-evaluated output. But the new proposed modified constant logic style and clocked logic style provides better result than the constant logic style. The CMOS technology is used for the simulation process by which the parameter of power is measured which is compared with constant delay logic style.

Keywords: *CMOS, power consumption, full adder, ripple carry adder, high performance, inverter, adder, constant delay*

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INTRODUCTION

CMOS is the most dominant technology which is used to construct the different types of integrated circuit. The ripple carry adder and full adders are generally used for defining the different logic styles. Full adder has three inputs and two outputs. A and B are the first two inputs of full adder and third input is input carry of the FA which is denoted by CIN. Output carry and Sum is the two output of the FA which is denoted by COUT and S respectively.

An inverter is consisting of nMOS and pMOS transistor. One end of the pMOS is connected to supply and one end of the nMOS is connected to ground. When input is '0' then nMOS transistor is OFF and the pMOS transistor is ON and output gets '1'. When input is '1' then nMOS is ON and pMOS is OFF and output gets '0'.

The ripple carry adder (RCA) is one of the simplest adders to implement. This adder takes in two N-bit inputs (where N is a positive integer) and produces (N+1) output bits (an N-bit sum and a 1-bit carryout). After adding the 'N' number of full adder together with the carry out bit of first full adder to the next carry in bit of full adder; then formation of ripple carry adder takes place. Ling's algorithm is used in energy efficient adder technique [1].

The energy optimized in the 64 bit gets the result of 240 ps delay by using the ling equation [2]. In 65 nm technology of ARM cortex-M0 up in which rail clamping technique is used with SCPG [3, 4]. The result of simulation produces the reduced power consumption [5] and clock frequency of 400 KHz [6, 7].

On 32:1 multiplexer, the design topology technique is proposed which is fabricated in 45 nm technology with 1 V supply [8, 9]. The result shows minimum current and leakage of 74 and 38% respectively [10, 11]. MSleep i.e. integrated share is helpful to reduce a power [12].

In nanometer technology, leakage power is reduced by using layout aware power rating methodology [13, 4]. Hspice method is useful for accuracy of model in CMOS technology [11].

CD LOGIC STYLE

The schematic circuit representation of CD logic is as shown in Figure 1 [14].

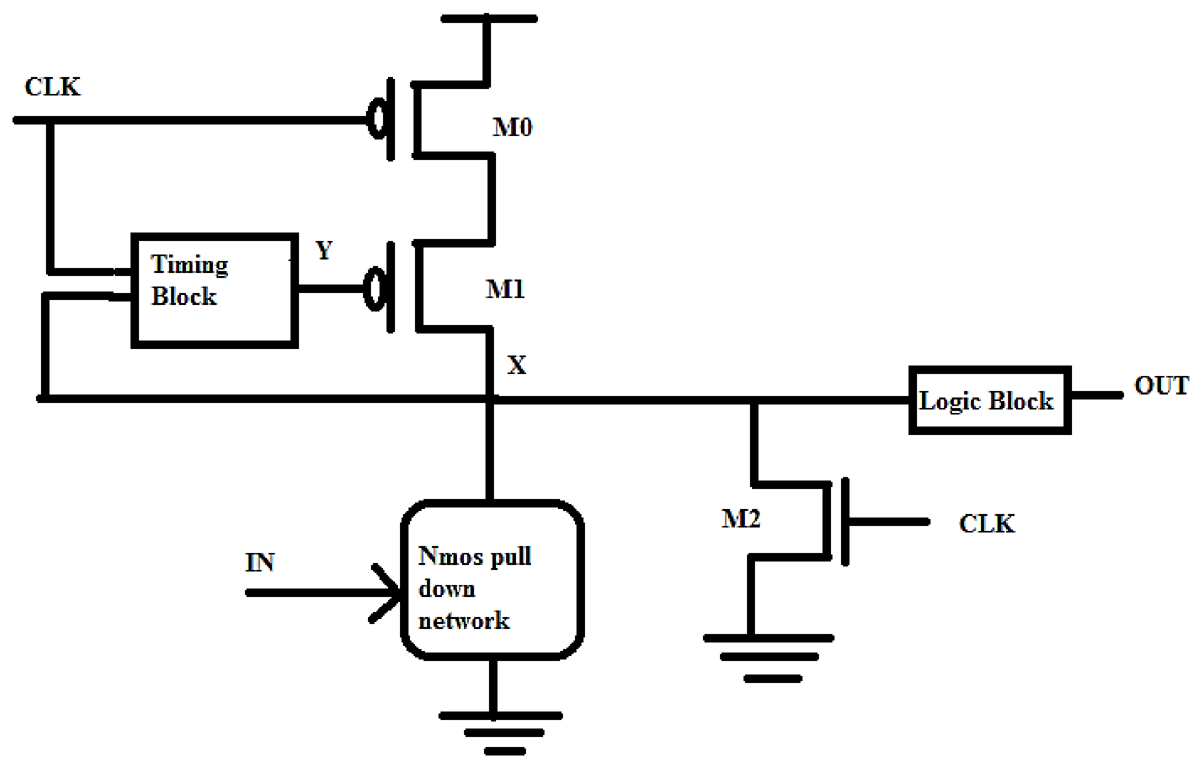

Fig. 1: *Constant Delay Logic Style.*

Constant delay logic style is having a unique characteristic for logical expression. Timing block and logic block, these are the two blocks of the constant delay logic style. The timing block is consisting of local adjusting window and self-reset technique. It is helpful for reducing the static power dissipation. The local adjusting window is representing a chain of inverter in CD logic style [15]. The self reset is representing as a NOR gate of CD logic style. NOR gate in which one of the input is intermediate node X and other is CLK. Two modes of operation of CD logic style are as follows:

1. Pre-discharge mode,
2. Evaluation mode.

When CLK=1 then occurrence of pre-discharge mode takes place and when CLK=0 then occurrence of evaluation mode takes place. X and OUT are pre-discharge and GND and VDD are pre-charge during the pre-discharge mode. The evaluation mode takes place in three different modes as follows:

1. Contention mode: When IN is high, CLK=0 then occurrence of contention modes takes place. The intermediate mode X is at non-zero voltage level then produces the temporary glitch due to this the input transits from high to low.
2. C-Q delay mode: When IN goes to low before CLK=0 then occurrence of C-Q delay mode takes place. The intermediate node X is at 1 then output discharges to VDD.
3. D-Q delay mode: When IN goes to low after CLK=0 then occurrence of D-Q delay mode takes place. The intermediate node X enters into the contention mode and its value set to 1.

METHODOLOGY

Modified CD Logic Style

Modified CD logic is having three changes than the CD logic style.

1. Window adjustment technique is eliminated.
2. Evaluation transistor is altered as pMOS transistor instead of nMOS.
3. Addition of transistor M2 and M3 in parallel below the pull down network.

M0 and M1 driven CLK and NOR gate which increases the resistance help to reduce power. M4 is acting as evaluation transistor. Power delay product is reduced by connecting M2 and M3 are in parallel below the pull down network.

In which M2 is helpful for reducing the power after increasing the values of resistance. Nor gate acts as a self rest circuit by the transistor M5, M6, M7, M8 which is driven by CLK and output intermediate node X. The schematic circuit representation of modified CD logic is as shown in Figure 2.
The two modes of operation are:
1. Precharge mode (CLK=0),
2. Evaluation mode (CLK=1).

When CLK=0, then occurrence of pre-charge mode takes place and when CLK=1, then occurrence of evaluation mode takes place. Transistor M4 is ON when CLK=0. Pull down network nMOS gets evaluated and output is produced then transistor M0 is OFF.

Due to this contention modes get converted into evaluation mode; due to this window adjustment technique get eliminated. Reduction in power and delay is due to the elimination of window adjustment technique. Transistor M2 and pull down network nMOS get ON at the time of evaluation mode and provide reduced power consumption. For this situation M3 is kept always 'ON' which is helpful for discharging the values to the ground.

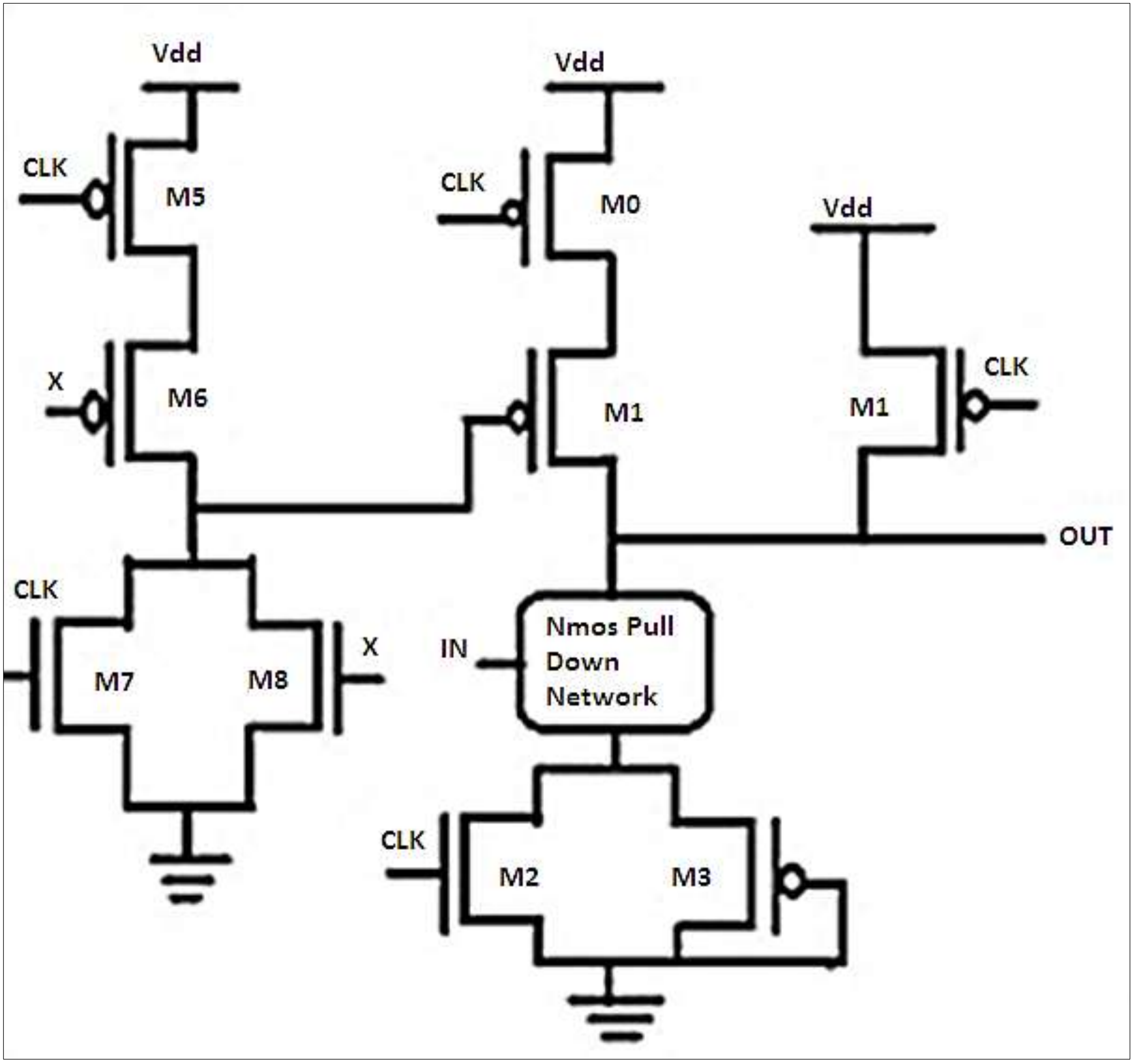

Fig. 2: *Modified Constant Logic Style.*

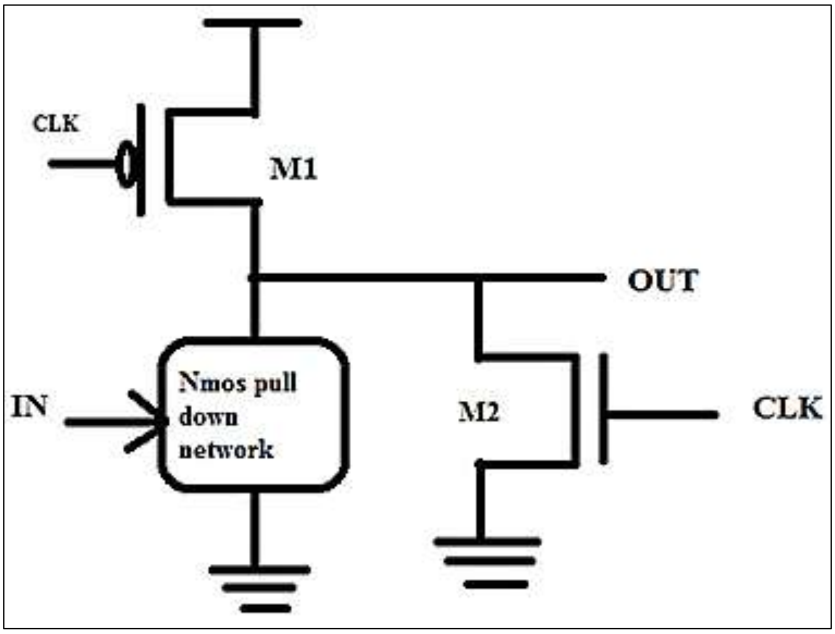

Fig. 3: *Clocked Logic Style.*

Clocked Logic Style

The basic operation of clocked logic style is shown in Figure 3.

When CLK=0, M1 is "ON" and M2 is "OFF", then the evaluation period begins. But when CLK=1, pre-discharged period begins and output is pulled down to ground through M2. "OUT" enters into the contention mode where transistor M1 and nMOS PDN are conducting current simultaneously. If PDN is "OFF" then output is rises to logic "1".

RESULT

The results obtained in the present study are given in tabulated form in Tables 1–3.

Table 1: *Inverter Performance Comparison.*

Parameter	CD Logic Style	MCD Logic Style	Clocked Logic Style
Power (mW)	20	13	5
Risetime	2.3195e–009	3.5641e–010	6.1894e–010
Falltime	1.4813e–010	3.1597e–011	4.5430e–011

Table 2: *FA Performance Comparison.*

Parameter	CD Logic Style	MCD Logic Style	Clocked Logic Style
Power (mW)	86	40	25
Risetime	1.5558e–010	1.666e–010	1.7102e–010
Falltime	1.3162e–011	1.2985e–011	1.3041e–011

Table 3: *RCA Performance Comparison.*

Parameter	CD Logic Style	MCD Logic Style	Clocked Logic Style
Power (mW)	114	45	38
Risetime	–1.0282e–008	–1.0273e–008	–1.0273e–008
Falltime	1.7911e–011	1.7912e–011	1.7913e–011

CONCLUSION

A new high performance modified constant delay logic style and clocked logic style was proposed. It will increase performance and reduce the power as compared to the constant delay logic style. Adders are designed using both existing as well as proposed logic. It is simulated in CMOS technologies for comparing performance parameter power, rise-time and fall-time. From the results it is found that MCD and clocked logic style is having better power than the existing logic style.

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